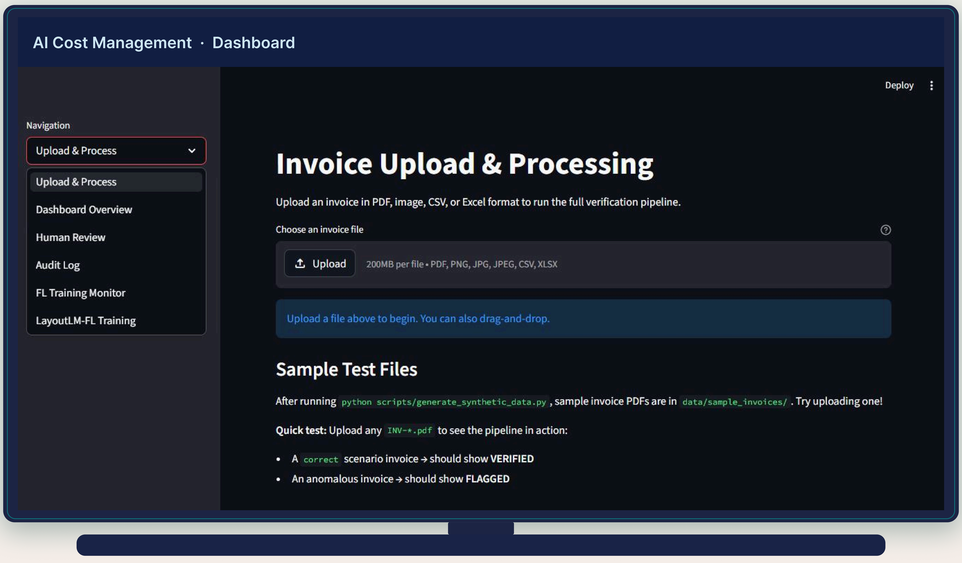
IBA Institute of Business Administration Karachi
Leadership and Ideas for Tomorrow

Herbion NATURALS

IBA SMCS School of Mathematics and Computer Science

AI-Based Cost Management System

AI-Powered Invoice Intelligence & Fraud Detection with Federated Learning + LoRA



Four-Tier Extraction Cascade & Pipeline

T1	FL-LoRA LayoutLMv3 - federated fine-tuned; highest accuracy on known invoice styles
T2	Base LayoutLMv3 - layout-aware NER without FL adapter; catches unseen layouts
T3	Spatial PDF Parser + Regex - pdfplumber with coordinate-aware field patterns
T4	Tesseract + GOT-OCR 2.0 - universal fallback; always produces output

WHY IT WORKS

- 87.9% F1 — only 5.4% below centralised
- 294K trainable params vs 110M
- Fully local SQLite storage, audit-ready
- LoRA communication: 1,152 KB/round vs 1.2 GB full fine-tune
- GDPR-safe by design, raw invoices are never centralised

Huzaifa Ahmed Khan • Muhammad Bilal Adnan • Ali Saeed
Supervisor: Dr. S.M. Faisal Iradat

h.khan.26485@khi.iba.edu.pk | BS Computer Science | FYP 2026 | Telecommunications Research Lab